

Mini Lesson Sample Script

Materials: Linking cubes in two colors, personal white board

1. Explore how adding the same amount to both numbers in a subtraction expression keeps the difference the same using linking cubes.

Build a stick of 7 cubes and a stick of 4 cubes.
Imagine each cube is worth 10. What is the difference?

Now add 1 cube to each stick. What is the new subtraction sentence? Did the difference change?

Add 1 more cube to each stick. What is the new subtraction sentence? Did the difference change?

Answer: $70 - 40 = 30$; $80 - 50 = 30$, the difference did not change; $90 - 60 = 30$, the difference did not change.

T: Here are two sticks of cubes. Each cube is worth 10. Let's count the longer one by tens.

S: 10, 20, 30, 40, 50, 60, 70.

T: And this one?

S: 10, 20, 30, 40.

T: What's the difference between 70 and 40? Say the subtraction sentence.

S: $70 - 40 = 30$.

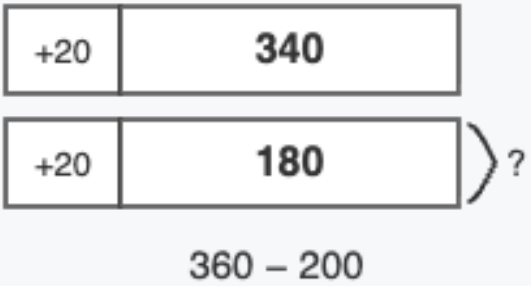
T: Watch what I do. I'm going to add 1 cube to each stick. What is the new subtraction sentence?

S: $80 - 50 = 30$.

T: Did the difference change?

S: No, it's still 30.

	T: Let me try one more. I'll add another cube to each stick. What's the new subtraction sentence now? S: $90 - 60 = 30$. T: And the difference? S: Still 30! T: As long as we add the same amount to both numbers, the difference stays the same. We can use that idea to make hard subtraction problems easier.
2. Subtract from a three-digit number by adding to both numbers to make a friendly multiple of ten.	
Solve $340 - 180$. Use a tape diagram to add the same amount to both numbers to make the problem easier, then write a new subtraction sentence and solve. <i>Answer: Add 20 to both numbers; $360 - 200 = 160$, so $340 - 180 = 160$.</i>	T: Let's try $340 - 180$. This one feels tricky. Look at 180. What friendly number is it close to? S: 200. T: How much do we need to add to 180 to make 200? S: 20.

 360 – 200	T: Draw two bars on your white board, one for 340 on top and one for 180 on the bottom. Now add 20 to each bar. S: (Draws the diagram and adds 20 to each bar.) T: What's the new subtraction sentence? S: 360 – 200. T: Solve it. S: 360 – 200 = 160. T: We added 20 to both numbers, so the difference didn't change. What does 340 – 180 equal? S: 160.
3. Subtract from a three-digit number by adding to both numbers to make a friendly hundred.	
Solve 423 – 198. Add the same amount to both numbers to make a friendly hundred, then write a new subtraction sentence and solve. <i>Answer: Add 2 to both numbers; 425 – 200 = 225, so 423 – 198 = 225.</i>	T: Here's another one: 423 – 198. Which friendly hundred is 198 close to? S: 200. T: How much do we need to add to 198 to make 200? S: 2.

T: Draw a tape diagram and add 2 to each bar.
Then write the new subtraction sentence.

S: (Draws the diagram.) $425 - 200$.

T: Solve it.

S: $425 - 200 = 225$.

T: We added 2 to both numbers, so the difference stays the same. So what is $423 - 198$?

S: 225.

T: Nice work. When you see a number close to a friendly ten or hundred, you can add the same amount to both numbers and make the subtraction much easier.